RAJAN KUMAR

CYBERSECURITY · AI / ML ENGINEERING · FRESHER

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PROFILE

Final-year Computer Science undergraduate (fresher) building a foundation across networks, cryptography and applied AI/ML. Completed project training at BSPTCL analysing a critical-infrastructure malware case, and an AI/ML internship at NIELIT. CCNA-trained; builds full-stack products and security tooling. Seeking a cybersecurity or AI engineering role.

EDUCATION

B.Tech, Computer Science & Engineering — Expected 2027

Sitamarhi Institute of Technology (Bihar Engineering University) | CGPA: 7.81 / 10

Class XII (Science): 67% | Class X: 65%

PROJECTS

PDG — Personal Digital Guardian — Python, FastAPI, PostgreSQL, React, Kotlin

- Privacy-first security companion across web, MV3 extension, Windows agent and Android. Deterministic, versioned rule engine scores risk with stored evidence; an optional LLM only narrates. Argon2id with rotating refresh tokens, STRIDE threat model, 9 design records, 299 passing API tests, Docker and Terraform deployment.

NetGuard AI — ML Network Intrusion Detection — Python, FastAPI, XGBoost, SHAP, Scapy, React

- Scapy capture to flow features to classifier plus anomaly model, with SHAP explanations, RBAC and tenant isolation, live WebSocket dashboard, human-approved retraining from analyst feedback and an HA deployment topology. Trained on synthetic data; real-dataset evaluation is pending.

AI Privacy Guardian — Python, FastAPI, React, TypeScript, Kotlin, Tauri

- Privacy platform for web, desktop, extension and Android: browser-side AES-GCM password vault (PBKDF2), k-anonymity breach check, local photo-metadata cleaner, AI scam and deepfake checks with offline fallback. 31 backend and 29 web tests.

PhishGuard — Privacy-Preserving Phishing Detector — JavaScript, Chrome MV3, Node.js, SQLite

- Extension scoring links locally with explainable heuristics; optional blocklist lookup sends only a 16-bit hash prefix; domain-age check is a separate opt-in. 8 passing test suites and an evaluation harness reporting false alarms per 1,000 links.

CyberShield — Threat Detection System — Python, Scapy, Flask, scikit-learn

- 8 rule-based detectors (port scans, SYN floods, brute force, DNS tunneling) mapped to MITRE ATT&CK, plus an Isolation Forest anomaly model; live Flask + Socket.IO dashboard with a REST API.

INTERNSHIP & TRAINING

Intern — AI and ML using Python, NIELIT Gorakhpur (Online) — Mar – May 2026 (90 hrs)

- Python, database connectivity, Tkinter GUIs, speech recognition, NumPy/Pandas/Matplotlib; core ML, deep learning and NLP concepts. Graded S (80%+).

Project Trainee — IT Department, BSPTCL, Patna — Jun 2025 (4 weeks)

- Project training on real-time case studies in critical-sector security, focused on the TRISIS malware attack (2017) on industrial safety instrumented systems.

SKILLS

Languages: Python (Advanced, 2+ yrs), JavaScript / TypeScript, Java / Kotlin (Intermediate), C, C++, HTML, CSS
Backend & Platform: FastAPI, Flask, Node.js, Express, PostgreSQL, SQLite, Redis, SQLAlchemy, Docker, Kubernetes, Terraform, GitHub Actions, pytest
Web & Client: React, Vite, Tailwind, Chrome MV3 extensions, Tauri, Android (Kotlin, Compose)
Security & Networking: CCNA (3-course track), Computer Networks, Cryptography, Threat modeling (STRIDE), MITRE ATT&CK, Scapy, Argon2id / JWT / TOTP auth
AI / ML: scikit-learn, XGBoost, SHAP, Isolation Forest, Deep Learning, NLP, NumPy, Pandas, LLM API integration
Core CS & Tools: DBMS, OS, DSA, Compiler Design, Blockchain, IoT, Linux, Windows

CERTIFICATIONS

- NPTEL (IIT Kharagpur): Blockchain and its Applications (Elite, 65%, 2026); Introduction to Internet of Things (58%, 2025)
- iHUB DivyaSampark, IIT Roorkee × Govt. of Bihar: Internet of Things, 40-hr program (Mar 2025)
- Cisco Networking Academy: CCNA (Introduction to Networks; Switching, Routing & Wireless Essentials; Enterprise Networking, Security & Automation); Python Essentials 1 & 2; C++ Essentials 1; Introduction to Modern AI; Introduction to Data Science; Find Insights with AI; Apply AI: Analyze Customer Reviews; Operating Systems Basics (with Linux, Windows and Mobile OS modules)
- NASSCOM × Skill India Digital Hub (1-hr participation, Oct 2026): IoT Security Analyst; Network Security Engineer